Adolescent milk fat and galactose consumption and testicular germ cell cancer.

Recent case-control studies suggested that dairy product consumption is an important risk factor for testicular cancer. We examined the association between consumption of dairy products, especially milk, milk fat, and galactose, and testicular cancer in a population-based case-control study including 269 case and 797 controls (response proportions of 76% and 46%, respectively). Dietary history was assessed by food frequency questions for the index persons and through their mothers including diet 1 year before interview and diet at age 17 years. We used conditional logistic regression to calculate odds ratios as estimates of the relative risk (RR), 95% confidence intervals (95% CI), and to control for social status and height. The RR of testicular cancer was 1.37 (95% CI, 1.12-1.68) per additional 20 servings of milk per month (each 200 mL) in adolescence. This elevated overall risk was mainly due to an increased risk for seminoma (RR, 1.66; 95% CI, 1.30-2.12) per additional 20 milk servings per month. The RR for seminoma was 1.30 (95% CI, 1.15-1.48) for each additional 200 g milk fat per month and was 2.01 (95% CI, 1.41-2.86) for each additional 200 g galactose per month during adolescence. Our results suggest that milk fat and/or galactose may explain the association between milk and dairy product consumption and seminomatous testicular cancer.